Project ID: 25-26J-426

Title of the Report:
Feasibility Study – Ml-based Mobile App for Early Risk Detection for Midwifery Support

Project Name:
VR & Mobile Application Solutions for Maternal Education

Student Name & ID:
Kumarage M.B.S.I.E. – IT22606556

Specialization:
Interactive Media

Submission Date:
11/01/2026

Executive Summary

This report evaluates the feasibility of developing a **Machine Learning–Based Interactive Mobile Application for Early Risk Detection in Midwifery Care**. The proposed solution is designed to assist Public Health Midwives by providing early prediction of maternal health risk levels using machine learning models integrated into a mobile platform.

The system analyzes maternal health indicators such as age, blood pressure, blood sugar level, heart rate, and body temperature to classify pregnancy risk levels (*Low, Medium, High*). Findings indicate that the project is **technically feasible** due to the availability of validated maternal health datasets and proven machine learning algorithms. **Market feasibility** is strong, as frontline healthcare workers lack digital decision-support tools. **Financial feasibility** is positive since the system is developed using open-source technologies. The study recommends proceeding with development, focusing on usability, model explainability, and real-world pilot testing.

.

Table of Contents

Executive Summary 2
Introduction 4
Background 4
Objective of the Feasibility Study 4
Scope of the Feasibility Study 4
Problem Statement 5
Project Description 5
Proposed Solution 5
Key Features Implemented 5
Technology Used 6
Feasibility Analysis 6
Technical Feasibility 6
Market Feasibility 6
Financial Feasibility 7
Operation Feasibility 7
Ethical and Environmental Feasibility 8
Risk Analysis 8
Recommendations 8
Conclusion 8
References 9

Introduction

Background

This report evaluates the feasibility of developing a **Machine Learning–Based Interactive Mobile Application for Early Risk Detection in Midwifery Care**. The proposed solution is designed to assist Public Health Midwives by providing early prediction of maternal health risk levels using machine learning models integrated into a mobile platform.

The system analyzes maternal health indicators such as age, blood pressure, blood sugar level, heart rate, and body temperature to classify pregnancy risk levels (*Low, Medium, High*). Findings indicate that the project is **technically feasible** due to the availability of validated maternal health datasets and proven machine learning algorithms. **Market feasibility** is strong, as frontline healthcare workers lack digital decision-support tools. **Financial feasibility** is positive since the system is developed using open-source technologies. The study recommends proceeding with development, focusing on usability, model explainability, and real-world pilot testing.

Objective of the Feasibility Study

The objective of this feasibility study is to assess the **technical, operational, and financial viability** of developing an AI-assisted mobile application that supports midwives in early maternal risk detection.

Scope of the Feasibility Study

This study evaluates the feasibility of:

- Machine learning model development and validation
- Mobile application integration
- Risk visualization and user interaction design
- Deployment in real-world midwifery settings

Problem Statement

The core problem addressed by this project is the **lack of real-time, data-driven decision-support tools** for midwives during routine maternal checkups.

- **Evidence:** Current maternal risk assessment relies on manual calculations and subjective judgment.
- **Significance:** Delayed or inaccurate identification of high-risk pregnancies can result in severe complications for both mother and fetus.

An automated risk prediction system can significantly improve early detection and clinical decision-making.

Project Description

Proposed Solution

The proposed solution is an **Interactive Mobile Application for Midwives** integrated with a machine learning–based maternal risk prediction model.

Key Features:

- Automated maternal risk classification (*Low, Medium, High*)
- Interactive visual indicators for risk interpretation
- Simple data input aligned with existing midwife workflows

Methodology: The system uses supervised machine learning models trained on a validated maternal health dataset referenced in prior academic research. The backend is developed using **Python**, while the mobile interface is designed for ease of use and clarity.

Key Features Implemented

- Idle, roaming, and wait behaviors
- Herd behavior (stick-with-herd logic)
- Hunger and thirst meters
- Drinking and grazing behavior
- Fleeing from player
- Chasing and attacking (depends on the species)
- Collision-aware movement
- 60+ average fps during runtime
- Separate PC build for more accessible testing

Technology Used

- Unity 6 (HDRP)
- C# scripting
- Hybrid system incorporating different behavior systems (State machines, GOAP, Utility systems)
- VR headset support (Meta Quest 3)

Feasibility Analysis

Technical Feasibility

- **Requirements:** Python (Scikit-learn), machine learning classification models, mobile application framework
- **Assessment:** The project is technically feasible, with high model accuracy achieved during initial testing
- **Challenges:** Model generalization and explainability
- **Mitigation:** Cross-validation techniques and explainable AI methods

Market Feasibility

- Target Users: Public Health Midwives and maternal clinics
- Demand: Strong demand for digital healthcare solutions that support early risk detection
- Relevance: Limited availability of AI-assisted maternal care tools in Sri Lanka

Financial Feasibility

- Development Costs: Low, as the system uses open-source tools and datasets
- Operational Costs: Minimal, requiring only standard mobile devices
- Return on Investment: Measured through improved maternal health outcomes and reduced complications

Operation Feasibility

- Designed to integrate into existing maternal care workflows
- Minimal training required for end users
- Scalable for future enhancements

Ethical and Environmental Feasibility

- Acts as a **decision-support tool**, not a replacement for medical professionals
- Ensures confidentiality and ethical handling of patient data
- Promotes safer maternal healthcare practices

Risk Analysis

Risk	Impact	Mitigation
Model performance degradation	Medium	Cross-validation, periodic model retraining
Input validation, predefined clinical ranges	Medium	Input Validation and Predefined clinical ranges
Mobile app performance lag	Low	Lightweight ML models, optimized inference
User resistance to adoption	Low	Simple UI/UX design and basic training

Recommendations

- Enhance explainable AI features for trust and transparency
- Conduct pilot testing with midwives
- Expand dataset with local clinical data
- Improve visual dashboards for better interpretation

Conclusion

The feasibility study confirms that the **Machine Learning–Based Interactive Mobile App for Early Risk Detection for Midwifery Support** is a viable and impactful project. By leveraging machine learning and mobile technology, the system can support midwives in identifying high-risk pregnancies early, improving maternal healthcare outcomes through a technically sound and cost-effective solution.

References

[1] Ahmed, M., et al. (2021). *IoT Based Risk Level Prediction Model for Maternal Health Care*. IEEE.

[2] UCI / Kaggle Maternal Health Risk Dataset.

[3] Rahman, A., & Alam, M. G. R. (2023). *Explainable AI based Maternal Health Risk Prediction*. IEEE.